$$QD = \frac{1}{2}(Q_3 - Q_1) = \frac{40-15}{2} = 12.5$$

Coefficient of QD

$$= \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{40-15}{40+15} = \frac{25}{55} = 0.455$$

Hence coefficient of QD $= 0.455$

Example 8.17

Compute coefficient of quartile deviation from the following data

Marks	10	20	30	40	50	60
No. of Students	4	7	15	8	7	2

Solution:

Marks	Frequency	Cumulative Frequency
X	f	cf
10	4	4
20	7	11
30	15	26
40	8	34
50	7	41
60	2	43
		$N = \sum f = 43$

Table : 8.11

$$Q_1 = \text{Size of } \left(\frac{N+1}{4}\right)^{th} \text{ value} = \text{Size } 11^{th} \text{ value} = 20$$

$$Q_3 = \text{Size of } \left(\frac{3(N+1)}{4}\right)^{th} \text{ value}$$

$$= \text{Size of } 33^{rd} \text{ value} = 40.$$

$$QD = \frac{1}{2}(Q_3 - Q_1) = \frac{40-20}{2} = 10$$

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{40-20}{40+20}$$

$$= \frac{20}{60} = 0.333$$

Example 8.18

Compute Quartile deviation from the following data

CI	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80
f	12	19	5	10	9	6	6

Solution:

CI	f	cf
10 – 20	12	12
20 – 30	19	31
30 – 40	5	36
40 – 50	10	46
50 – 60	9	55
60 – 70	6	61
70 – 80	6	67
	N = 67	

Table : 8.12

$$Q_1 = \text{Size of } \left(\frac{N}{4}\right)^{th} \text{ value} = \left(\frac{67}{4}\right)^{th} = 16.75^{th} \text{ value}$$

Thus Q_1 lies in the class 20 – 30; and the corresponding values are

$$L = 20,\ \frac{N}{4} = 16.75,\ pcf = 12,\ f = 19,\ c = 10$$

$$Q_1 = L + \left(\frac{\frac{N}{4} - pcf}{f}\right) \times c$$

$$Q_1 = 20 + \left(\frac{16.75 - 12}{19}\right) \times 10 = 20 + 2.5 = 22.5$$

$$Q_3 = \text{Size of } \left(\frac{3N}{4}\right)^{th} \text{ value} = 50.25^{th} \text{ value}$$

Thus Q_3 lies in the class 50 – 60 and the corresponding values are

$$L = 50,\ \frac{3N}{4} = 50.25,\ pcf = 46,\ f = 9,\ c = 10$$

$$Q_3 = L + \left(\frac{\frac{3N}{4} - pcf}{f}\right) \times c$$

$$Q_3 = 50 + \left[\frac{50.25 - 46}{9}\right] \times 10 = 54.72$$

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